NSF Data Science REU at HMC

Analysis of Time Course RNA-Sequencing Data in *T. brucei* and *E. coli*

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Bromodomain Proteins and Development in *T. brucei*

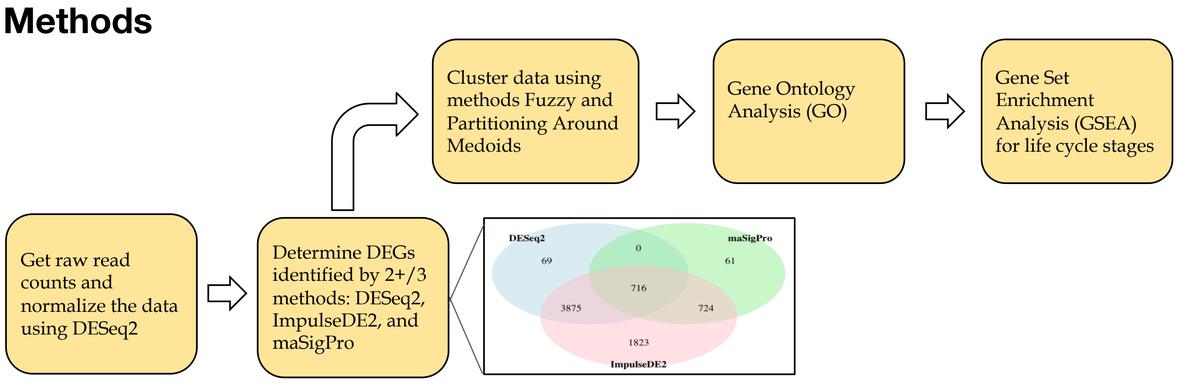
Introduction

West African Trypanosomiasis, most commonly known as sleeping sickness, is a disease with devastating health and economic consequences: several hundred thousand infections occur annually and if left untreated, it can result in death (Legros *et al.* 2002). Infections are transmitted by the bite of a Tsetse fly carrying the protozoan parasite *Trypanosoma brucei* (*T. brucei*). During its life cycle, the parasite switches between procyclic form (PF) and bloodstream form (BF). In *T.brucei* BF, the parasite becomes densely covered with highly immunogenic variant surface glycoproteins (VSGs), allowing it to evade the mammalian antibody immune response (Savage *et al.* 2012).

In mammals, histone acetyl-lysine recognition by bromodomain proteins are known to control stem cell fate (Di Miccio *et al.* 2014). Previous work showed that bromodomain proteins are responsible for the maintenance of *T. brucei* BF stage identity (Schulz *et al.* 2015).

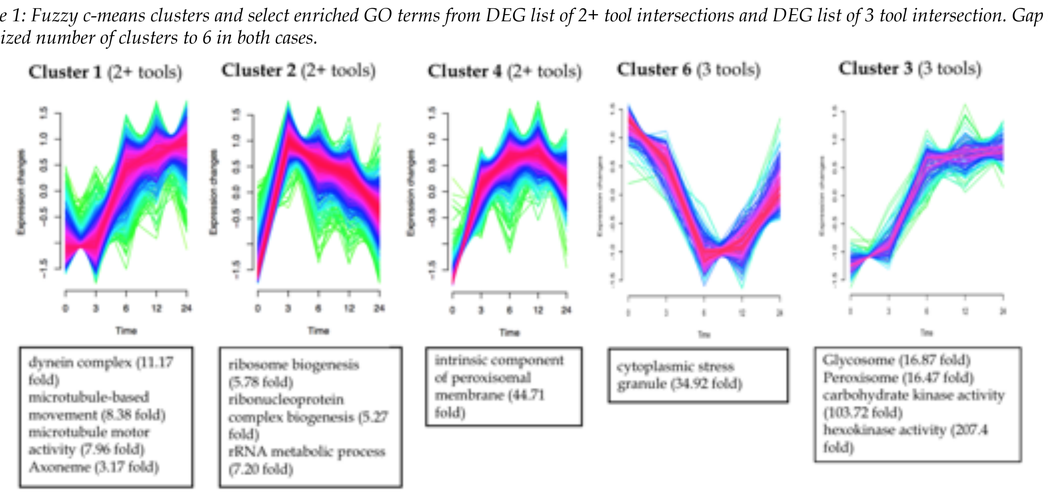
To determine the role of bromodomain proteins in PF *T. brucei*, an RNA-Seq experiment measuring gene expression in WT PF and bromodomain-inhibited PF over a 2 week time course was conducted. Results were compared against known transcriptomic profiles of life-cycle stages from Savage et al (Savage *et al. 2015).*

Methods



Results

Figure 1: Fuzzy c-means clusters and select enriched GO terms from DEG list of 2+ tool intersections and DEG list of 3 tool intersection. Gap statistic optimized number of clusters to 6 in both cases.



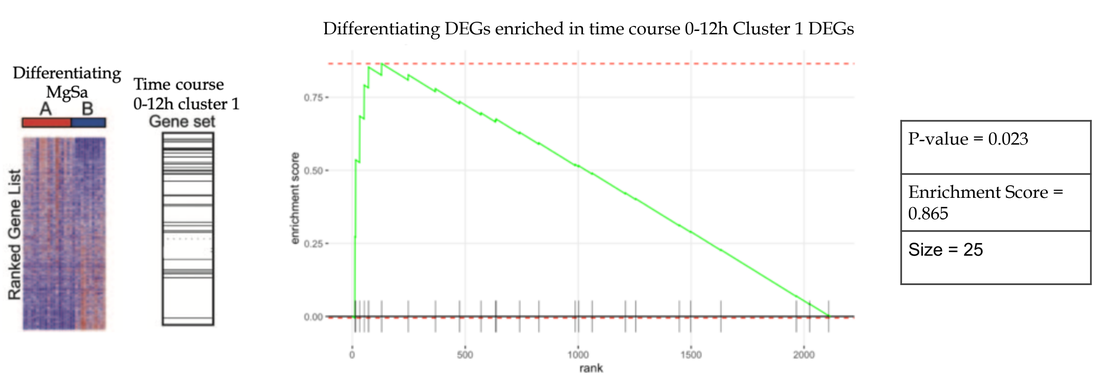


Figure 2: Gene Set Enrichment – Differentiating DEGs enriched in Schulz 0-12 hours cluster 1 DEGs

Conclusions and Future Directions

- **GO analysis of Fuzzy clusters suggest bromodomain proteins control processes related to motility, glycosomal metabolism, and maintenance of cytosolic stress granules.**
  - **Motility**: Cluster 1 GO terms indicate upregulation of motility related genes. *T. brucei* movement through host tissues is essential for development & pathogenesis, particularly in migrating through the tissues of the insect host (Dimitrova *et al.* 2017).
  - **Glycosome & hexokinase activity**: Cluster 3 GO terms indicate upregulation of glycolic and glycosomal genes. The bloodstream is a glucose-rich environment; in BF, glycolysis is confined to the glycosome and is the sole source of ATP (Michels *et al.* 2006).
  - **Stress Granule**: Cluster 6 GO terms indicate downregulation in genes involved in the cytoplasmic stress granules. These granules are RNA-protein complexes formed during Carbon-starvation in the insect intestinal tract (PF).
- **Bromodomain proteins appear to play role in developmental programme in PF.**
- **Future directions:**
  - Recluster and run GO analysis to mine data for biological meaning
  - GSEA: look for clusters enriched in known differentiation DEGs to see which elements of the developmental programme are turned on
  - Prioritize sampling between 0-24 hrs for higher resolution of expression kinetics.
  - Determine if transcriptomic changes are accompanied by proteomic changes on cell surface, stress granules, and in glycosome.

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Timing of the RpoS regulon in *E. coli*

Introduction

Escherichia coli (E. coli) possesses a general stress response that coordinates physiological responses to a stressful stimuli such as cell starvation (Battesti *et al.* 2011, Hengge 2011). RpoS, a transcription factor involved in this response, regulates expression of one quarter of the *E. coli* genome (preliminary data from Prof Dan Stoebel).

Wong and colleagues previously proposed that different genes' transcriptional sensitivities to RpoS could control the *timing* of gene expression in response to stress *in vivo* (Wong *et al.* 2017). RpoS sensitivities were previously classified in three groups shown in *Figure 4*: *linear* genes are expressed linearly with RpoS concentration, *sensitive* genes are transcribed dramatically at low RpoS levels, and *insensitive* genes are transcribed very little at low RpoS levels.

Gene expression data was collected in a 150-minute cell starvation time course (TC) RNA-Seq experiment for WT & RpoS-knockout strains, and analyzed to ascertain whether RpoS-sensitivity accounts for differences in expression timing under cell starvation.

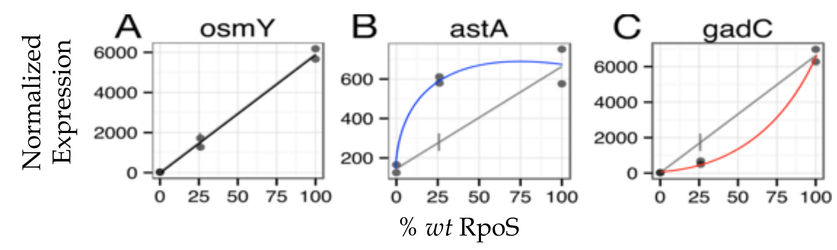
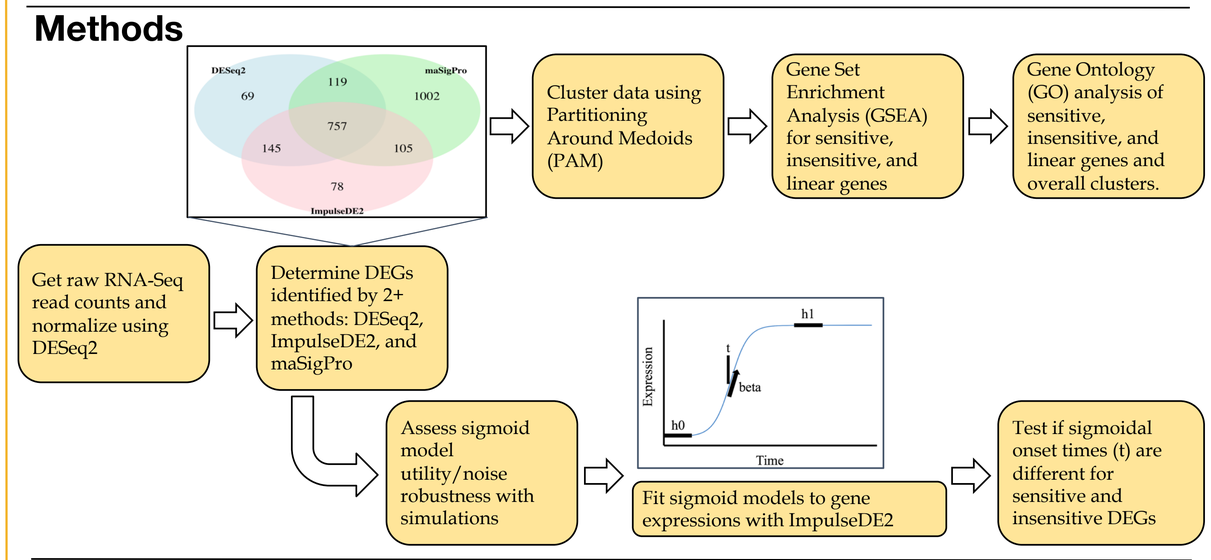


Figure 4: (A) expression profile of a linear gene: osmY (B) expression profile of sensitive gene: astA C) expression profile of an insensitive gene: gadC. Adapted from Wong et al. 2017.

Methods



Results

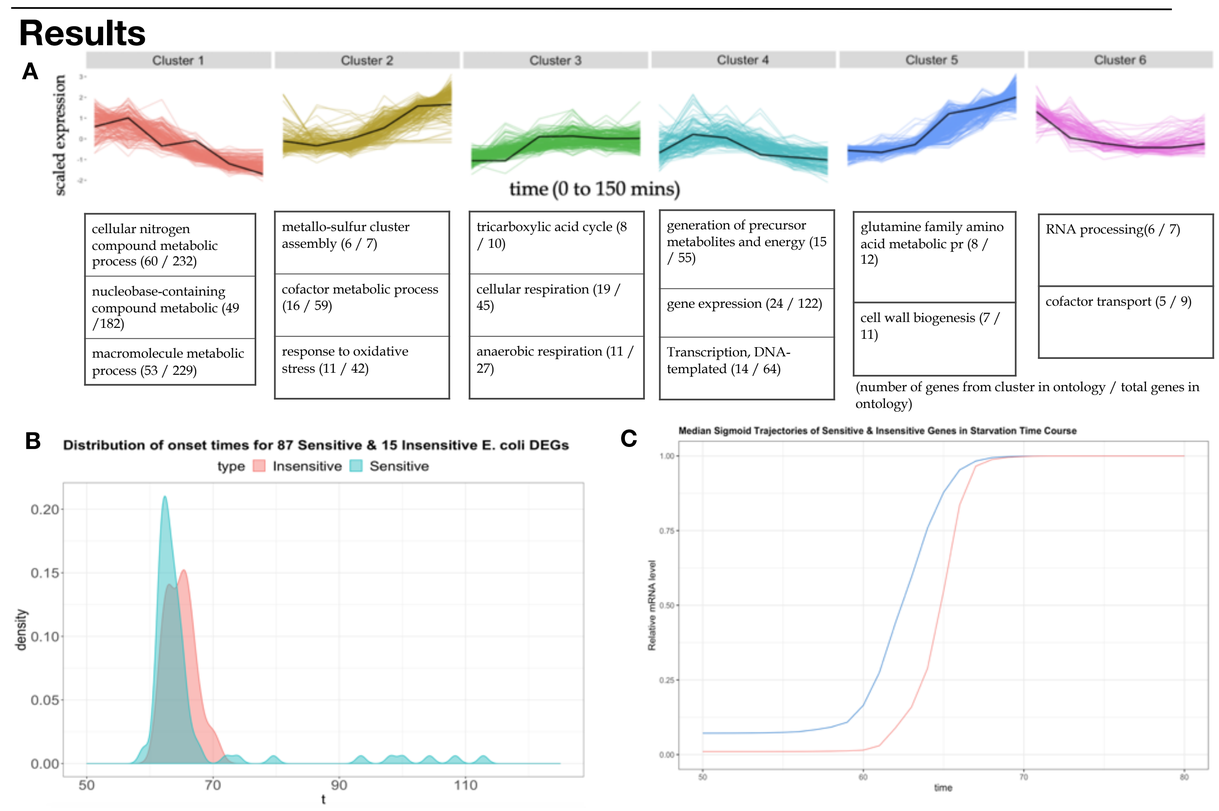


Figure 5: (A) Log2-transformed gene expression of medoid gene for each PAM cluster over time with corresponding cluster GO terms below (B) Distribution of sigmoidal onset times for monotonic sensitive and insensitive DEGs (C) Median sigmoid model for sensitive and insensitive relative expressions between 50 and 80 mins

Conclusions and Future Directions

- **Gene set enrichment analysis of sensitive and insensitive genes in PAM clusters provided insignificant enrichment results.**
- **GO analysis of PAM clusters** identified upregulation and downregulation of biologically significant gene ontologies.
- **RpoS sensitivity does not explain the overall trend in transcriptomic timing in *E. coli* response to cell starvation**
  - While a significant difference in trimmed onset times between sensitive and insensitive genes (Wilcoxon test p-value=0.009) was observed, RpoS-sensitivity likely does not strictly control the global dynamics of the transcriptomic response to cell starvation.
- Simulations demonstrated the utility and noise & outlier robustness of the sigmoid model's onset time parameter (t).
- **Future directions**:
  - Future TC RNA-seq experiments should increase sampling within the 55-75 min time frame for better temporal resolution of expression onset and timing cascades.
  - We recommend analysis of GO hierarchy to identify ontologies that turn on significantly earlier or later than expected.
  - Clustering of sigmoid models will prevent clustering to noise

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Acknowledgments

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